

Retrieval-Augmented Generation over arXiv Graph Neural Network Papers for Drug Discovery

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1. Introduction

When an LLM does not know certain information, it can be known to hallucinate. By building a RAG, it can ground an LLM in external documents to ensure that the LLM pulls the correct information when possible and says “I don’t know” instead of hallucinating. The goal of this project is to see if an LLM can effectively use a RAG to improve efficiency and accuracy. I found that BM25 outperformed dense retrieval on my selected corpus, and the system is highly faithful to retrieved context but occasionally over-answers unanswerable questions.

2. Corpus Construction

2.1 Source and Collection

The corpus was collected from the arXiv API, querying for papers combining graph neural network terminology with drug-discovery-related terms (drug discovery, molecular property, drug-drug interaction, molecule generation, virtual screening), restricted to the categories cs.LG, q-bio.QM, q-bio.BM, and cs.AI. Results were sorted by submission date, descending, up to a cap of 800 papers. The query returned 360 unique papers.

Category and date distribution:

Category	Count
cs.LG	263
q-bio.BM	35
q-bio.QM	19
physics.chem-ph	15
stat.ML	6
cs.AI	6
other	16

Publication dates range from 2017-09-12 to 2026-07-28, with a strong skew toward recent (2025-2026) papers.

I selected this niche for the corpus because I stay up to date with new clinical trials and thought that my existing knowledge may benefit me with this corpus. Categories and keywords were selected based on the clear buckets that were observed when looking at arXiv.

2.2 Preprocessing and Chunking

For each paper, the title and abstract were concatenated and used as the document text (full PDF text was not used, see limitations). Each document was assigned a unique ID equal to its arXiv identifier. Text was cleaned with whitespace normalization and split into word-based chunks of 200 words with a 40-word overlap. Additionally, documents shorter than 200 words were kept as a single chunk. Every chunk retains its parent document's ID, title, authors, publication date, category, and source URL for citation and traceability. This process produced 526 chunks from the 360 papers; 166 papers were long enough to be split into 2 or more chunks.

I chose to only use abstracts rather than full paper text because it allowed me to simplify the pipeline. Although this results in shallower context, the RAG is better trained off the keywords from abstracts as opposed to repeated content found throughout the papers.

2.3 Corpus Suitability Diagnostic

To verify the corpus was not already known to the generation LLM, 10 factual questions were written directly from the corpus content and put to the LLM with no retrieved context.

Results:

Outcome	Count	Example
Correctly declined ("I don't know")	4 / 10	CondPSE accuracy, MEGA-CL HLMC%, EnsembleEGNN R ² , FracAug AUROC gain
Confidently hallucinated wrong answer	4 / 10	Invented "Decagon" instead of EmerGNN; fabricated GENN PR-AUC numbers (14.6%/22.1% vs. actual 13.77%/5.01%)
Partial / generic-knowledge guess	2 / 10	Guessed path_boost was on PyPI (correct by default assumption); partially named synthesizability filters from general chemistry knowledge
Fully correct	0 / 10	—

With multiple strong hallucinations coupled with 0 correct answers, I concluded that this corpus would be effective since it clearly has not been memorized by the LLM.

3. System Design

3.1 Retrieval

- Classical: BM25 (rank-bm25 library), lowercase + regex tokenization (alphanumeric only), Okapi BM25 scoring, top-k = 5.
- Dense: sentence-transformers/all-MiniLM-L6-v2, L2-normalized embeddings, cosine similarity, top-k = 5. Embeddings are precomputed once and cached to disk.

3.2 Generation

Generation uses the Groq API running Llama 3.3 70B (llama-3.3-70b-versatile), temperature 0, a fixed seed (42) for reproducibility. The system prompt instructs the model to:

- Answer using only the retrieved context, no outside knowledge.
- Cite every claim inline with the source chunk ID, e.g. [chunk_id].
- Respond exactly "I don't know" (with a brief note on what's missing) when the context is insufficient.

The same model, temperature, and prompt are used for both the BM25 and dense retrieval conditions, so the two systems are compared on retrieval quality alone.

4. Evaluation Set Construction

4.1 Methodology

The gold evaluation set consists of 10 questions written by reading the corpus directly: 4 factoid, 3 multi-hop (each requiring synthesis across two source papers), and 3 unanswerable. Each question has a reference answer and the ground-truth source chunk ID(s) it was written from.

To create these questions, I used Claude AI to point me towards interesting topics that can be measured, then constructed questions after reading the recommended sections.

4.2 Potential Sources of Bias

- Topic skew: gold questions are concentrated on drug-drug interaction (DDI) papers, which are well-represented in the corpus but not necessarily representative of all 360 papers/topics covered.
- Small sample size (n=10) limits the statistical confidence of any reported metric.
- Single-annotator grading (no second team member cross-checked every human-graded answer) means there's no inter-rater reliability check.

5. Experimental Results

5.1 Retrieval Evaluation

Computed over the 7 factoid + multi-hop questions (unanswerable questions have no ground-truth chunk, so are excluded from retrieval scoring):

Retriever	MRR	Recall@5	Hit@5
BM25	1.00	0.93	1.00
Dense (MiniLM)	0.57	0.64	0.71

5.2 Generation Evaluation

Computed via human grading of all 20 generated answers (10 questions $\times$ 2 retrievers):

Retriever	Answer correctness	Citation faithfulness	Appropriate "I don't know" rate
BM25	80%	100%	67%
Dense (MiniLM)	60%	100%	67%

BM25 clearly outperforms dense on both retrieval and correctness. I believe this is because many questions I wrote rely on exact numbers and model names to be answered properly. Citation faithfulness is 100% for both retrievers, which makes me confident that there is little risk regarding hallucinations. However, the appropriate-IDK-rate is not 100% for either due to a wrong but similar idea being present in another document within the corpus. The retrievers did not hallucinate the answer, they rather misunderstood what the question was asking and instead produced answers that still appeared correct with existing context.

6. Error Analysis

6.1 Case: Near-miss unanswerable question (both retrievers)

Question g9 (“What clinical trial results does the MEGA-CL paper report for its ADMET predictions?”) was designed to be unanswerable, since the MEGA-CL paper reports preclinical/in vitro assay results, not clinical trial results. Both BM25 and dense retrieval correctly surfaced the MEGA-CL chunk (it is the most topically relevant document in the corpus), and the model answered with the true preclinical details from that chunk rather than recognizing that “clinical trial” and “preclinical in vitro assay” are not the same thing, and declining.

6.2 Case: Multi-hop retrieval failure (both retrievers)

Question g7 (“What two different strategies do these corpus papers use to improve GNN robustness under limited or out-of-distribution data?”) required retrieving two specific papers. BM25 found only 1 of 2 (Recall@5 = 0.5) and dense retrieval found 0 of 2 (Recall@5 = 0.0). Both systems substituted a different, real-but-incorrect paper into the answer instead.

The retrievers likely failed here because the query’s phrasing does not share exact vocabulary with either target abstract. Potential solutions to improve this in the future would be to retrieve more chunks per hop, or to have more specifications within the query.

6.3 Positive finding: graceful degradation under retrieval failure

On question g2, dense retrieval completely missed the correct chunk (Recall@5 = 0), yet the model correctly responded “I don't know” rather than fabricating an answer from irrelevant retrieved chunks. Similarly, on g6, dense retrieval found only one of the two required chunks, and the model answered the supported half correctly while explicitly declining the unsupported half rather than guessing.

This is important because it means that the LLM is effectively following its instructions to reliably prevent fabrication of information. It attempts to answer what it believes it knows, but it will not create false information just to produce an incorrect answer.

7. Strengths and Limitations

Strengths

- A citation faithfulness of 100% across all graded answers shows that the system never fabricates content beyond what it retrieves
- Factoid performance was strong, considering that the corpus is not memorized by the LLM
- The retrievers tend to decline answering rather than hallucinating information

Limitations

- The corpus only uses abstracts, which could limit the depth of answerable questions
- Only one embedding model and one LLM were tested, results may not generalize to other models
- The system struggles to distinguish topically related but insufficient evidence from truly supporting evidence

8. Reproducibility

<https://github.com/OMacgowan/rag-gnn-drug-discovery>

Demo video has been submitted as a direct file upload alongside this report.

Command to reproduce my findings: `python run_all.py`